

Gradient Computation

The gradient reflects the maximum slope of the gravity disturbance change.

The steepest parts of gravity anomalies defines the structures (width of anomaly) which decay with length

This step is made using QGIS.

- Load the .tif file of the gravity field

- Reproject coordinate in UTM

Raster / Projection / Warp (Reproject) :

- Target CRS : EPSG:32605 for Hawaii and EPSG:32628 for Tenerife
- Resampling method to use : Cubic (otherwise weird lines in gradient)
- Output data type : Float32

Run, change colorscale and then save the temporary layer.

- Compute the gradient

Select UTM Gravity layer

Raster / Analysis / Slope :

- Can check the box: compute edges

Put colorscale and change min and max if needed. Save.

- Get maximum gradient values

Create a new shapefile layer (line) and draw the axis of the ridge.

Vector / reproject layer - in UTM so values in m directly

Open the toolbox Points along geometry and create points on the line every 0.5 km.

Open the toolbox Geometry by expression. Select the points as input layer, line as output. In expression, copy that formula (with wanted length, and name of axis layer) :

For Tenerife:

```
make_line(  
  project($geometry, 10000,  
    radians(angle_at_vertex(geometry(get_feature_by_id('NE_riftaxis_UTM', 1)), 0) + 90)),  
  project($geometry, 10000,  
    radians(angle_at_vertex(geometry(get_feature_by_id('NE_riftaxis_UTM', 1)), 0) - 90))  
)
```

For Hawaii (because it is curved):

```
make_line(  
  project(  
    $geometry,  
    10050,
```

```

        azimuth(
            line_interpolate_point(geometry(get_feature('Rift_axis', 'id', 1)),
line_locate_point(geometry(get_feature('Rift_axis', 'id', 1)), $geometry) - 1),
            line_interpolate_point(geometry(get_feature('Rift_axis', 'id', 1)),
line_locate_point(geometry(get_feature('Rift_axis', 'id', 1)), $geometry) + 1)
        ) + radians(90)
    ),
    project(
        $geometry,
        10050,
        azimuth(
            line_interpolate_point(geometry(get_feature('Rift_axis', 'id', 1)),
line_locate_point(geometry(get_feature('Rift_axis', 'id', 1)), $geometry) - 1),
            line_interpolate_point(geometry(get_feature('Rift_axis', 'id', 1)),
line_locate_point(geometry(get_feature('Rift_axis', 'id', 1)), $geometry) + 1)
        ) - radians(90)
    )
)
)

```

- Convert the lines into points every 200m with Densify by interval then by Extract vertices.
- Inject the gradient values in the points : Set M value from raster, select your point layer and gradient layer
- See the gradient values in the attribute table : field calculator - decimal number - put in expression : `m($geometry)`

Save as csv file

download the topo for hawaii :

Install plugin : OpenTopography DEM Downloader

open it and select STRM 30m, zoom on your zone, API key needed.

And put hillshade, reproject topo in UTM the same way. HILLSHADE AND Z SCALE FACTOR 1 if in UTM and 0.0001 if in DD.

Next step in python

In python open the csv files and run the code to plot the distance.

So 51 points on both sides of the rift : = 10200 m so rift middle is 10100 m

So 0 - 50 is one side and 51 - 101 is the other side of the rift.

The code takes the point on each side where the gradient is maximum and converts it in distance from the center of the rift axis. In that way we obtain the plot of the width of the rift given its length.

Apply offset so start at the same point : caldera rim

0 for tenerife and 15 km for hawaii.

